VERIFY: what only the prototype can settle

Every value below was taken from a datasheet, a seller listing or a model. Nobody has measured it yet. Each item says how to check it and what to change if it is off. Work top to bottom: the first two sections gate the rest. Tick an item by noting the measured value and the date here.

SPICE predictions (`docs/verification/spice-report_main-v0.1_led-v0.2_rev2.md` and `.pdf`, rev 2, 2026-11-08) for several items: E12 is predicted to fail its 0.3 V limit (0.64 V dip for ~21 ms; R5 4.7k shortens it, does not reduce it); E13's hot-plug step is predicted to reset the MCU at 3.5 V (VSYS ~2.3 V; the step has been removed, D2), the power-on and empty-sleep steps to pass (VLED 10-90 % in ~1.0 ms, VSYS dip ~12 mV); E15 passes on a 5.0 V source (4.00 V) but not at 4.75 V through a 0.2 Ω cable with F1 at its 0.2 Ω maximum (3.71 V). Record the measured value beside the prediction, and update `sim/` with it.

1. Before the first power-up (bench, meter, no battery)

#	Item	How	Pass	If not
E1	Battery lead polarity	Meter on the cell's JST-PH plug: red = J2 pin 1 (silk "+")	+ on pin 1	Re-pin the plug (PH crimps lift out with a needle). Never "try it": with USB present a reversed cell is not blocked (round1 m2)
E2	SW1 ON direction (N5)	No battery; bench supply 3.7 V, 100 mA limit, on J2 (+ = pin 1). Slide SW1 both ways and meter TP3 (VSYS) to TP4 (GND)	One direction gives VSYS ≈ 3.7 V	Mark ON on the pack with a dot. If the "ON" side is the awkward one, flip `pack_front` text, not the PCB
E3	Supply sequencing	Bench 3.7 V on J2 (current limit 300 mA), switch ON, MCU flashed, nothing on J3	I < 5 mA; VLED (J3 pin 2) = VSYS once the firmware runs, 0 V while the MCU is held in reset	Check Q2/Q4 orientation and R6/R12
E4	Off current (R17)	Switch OFF, 4.0 V on J2, µA meter in series	≤ 10 µA	Lift Q2 / D3 to find the leak
E5	10 µF 0805 DC bias (N3)	LCR meter at 1 kHz with 4 V bias, or the charger ripple on a scope	≥ 4.7 µF effective	Replace C1/C2 with 1206 or a 25 V 0805
E6	D1 land pattern (N4)	Visual after reflow: fillets on both terminals	Solder fillet both ends	Hand-touch with the iron; widen the pads in v0.2

2. Firmware bring-up (handoff 04)

#	Item	How	Pass	If not
F1	Fuses written (FW-09)	`pio run -e attiny3216 -t fuses`, then flash the debug build and read the boot line	No "BOD OFF" line	Re-run the fuses target; check the programmer wiring
F2	Silicon revision (FW-12)	Debug build prints `SYSCFG.REVID` (add one line), or read it with the programmer	Any revision: the RTC is enabled for the PIT anyway	—
F3	VDD reading (m-9, FW-08)	Debug serial `V=` vs a meter on VSYS at 3.5, 3.8, 4.1 V (bench supply on J2)	Within ±50 mV	Add a calibration factor (measured / read) to `config.h` and scale `vdd_mv()`

#	Item	How	Pass	If not
F4	Empty sleep and resume	Bench supply on J2 down to 3.25 V for 6 s → fades out and sleeps. Then measure the sleep current. Raise it to 3.8 V or plug in USB	Sleep ≤ 30 µA (LED board unpowered); wakes in 3–5 s	Check PB2/Q4; check the PIT wake (F2)
F5	LED-board recovery (m6)	Unplug and replug J3 with the lamp on	Light returns within ~3 s	Debug log: `lp=` must return to `1C` and `ts=` to 1
F6	Button feel	Press, hold 2 s, auto-repeat	Matches R5–R8	Tune `DEBOUNCE_MS`, `REPEAT_*` in `lightcore.h`

3. Electrical characterisation

#	Item	How	Pass	If not
E7	LP5012 + MCP9800 supply current (LED board v0.2)	VLED 4.20 V, current into J3 pin 2: (a) all outputs at 0 for > 30 ms (power save), (b) one output on	(a) ≤ 0.45 mA (LP5012 ≤ 10 µA + MCP9800 ≤ 0.4 mA max, 0.2 mA typ; the idle bus draws nothing through the 3.3 k pull-ups), (b) ≤ 6 mA + the LED current	Recompute the runtimes with the measured value
E8	Channel current (RIREF 2.94 kΩ) and colour polarity	1 Ω in series with one white LED: colour 255 × brightness 255, then colour 0 × brightness 255 (the datasheet's OUTx_COLOR tables say FFh = 0 %, contradicting the rest of it: round3 R3-F1)	25.0 mA ± 5 % at colour 255, 0 mA at colour 0	Check R1 (0402 on the LED board's bottom); update `I_CHANNEL_MA` *and* the tests
E9	G/B regulation (M1, F10; v0.2)	First measure the received G and B dies' V_F at 25 mA, **and one 2700 K and one 5000 K white** (SPICE D4: worst-case whites start to dim just below the 3.45 V cap; if a white reads > ~2.85 V at 25 mA, raise `VBAT_LOW_MV` to 3500) (bench supply + 1 Ω, D9 on a spare board or before reflow). Then Green mode at max, VLED swept 4.2 → 3.4 V; scope across 1 Ω in series with the G die and read the **peak**, not the average (PWM). The LP5012 has no per-channel current step, so G/B always run at the 25 mA peak	Peak 25 mA held down to VLED ≈ V_F(measured) + 0.35–0.45 V (typical die ≈ 3.65 V; a maximum-V_F die (~3.7 V at 25 mA) needs ~4.05–4.15 V, i.e. it is never fully regulated: round3 R3-E3)	Accept (G/B dim gracefully below that), or RIREF 3.65 kΩ (20 mA for every channel)
E10	Charge current	USB in, cell at ~3.7 V: current into J2	256 mA ± 10 %	Check R3 3.9 k
E11	USB current draw (M4)	USB meter, charging + 4000 K full	Note the value (expect ~0.5–0.6 A)	Use a ≥ 1 A charger (handoff 02)
E12	Unplug handover (m1)	Scope VSYS while pulling USB at 3.6 V battery	Dip ≤ 0.3 V, no reset	Lower R5 further (4.7 k)
E13	Q4 soft turn-on (round2 R2-E1)	Scope VSYS (TP3) and VLED (TP7) at power-on and on the empty-sleep wake, at 3.5 V battery. J3 is not hot-plugged: it is mated once at assembly with SW1 off (SPICE D2, Matt 2026-11-08)	VLED ramps over ~1–2 ms; VSYS dips ≤ 0.2 V; no reset	Raise C8 (220 nF) or add 47 µF bulk on VSYS

#	Item	How	Pass	If not
E14	F1 PTC when warm (round2 R2-E3)	USB in, charging, 4000 K full for 30 min, pack closed; contact thermometer on F1	No trip; F1 ≤ 60 °C	Move F1 away from U1/D3 in v0.2
E15	VSYS on USB (round2 note)	USB in from a 5.0 V, ≥ 1 A USB-C charger, 4000 K full: meter TP3; then let the lamp reach empty sleep and plug the same charger in (SPICE D3, Matt 2026-11-08: the limit applies to a 5.0 V source; weak sources are expected to read lower)	VSYS ≥ 3.9 V on the 5.0 V source; wakes from empty sleep on USB. Informational: repeat the wake check on the weakest charger Matt actually uses (laptop port, hub); a late wake there is what triggers the remedy (D3 option B)	Lower `VBAT_RESUME_MV` to 3600 in `lightcore.h`
E16	LP5012 WQFN-20 joints (LED board v0.2)	After both reflows: diode-test every LED from the exposed end of its cathode pad (handoff 05 §2.3; the vias are tented); then power up and read DEVICE_CONFIG1 (debug log)	Every LED glows at the diode test and lights in its mode at first light; the read-back is 0x1C (`lp=1C`)	Dark at the diode test = LED reversed or open on top. Dark only at first light = an open OUTx joint under U1: reflow with flux. No I2C answer at all (`lp=--`) = EP (GND), VCC or SDA/SCL joint
E17	Thermostat path (v0.2)	Flash `bench_tstat_low` (TSET 30 °C / THYST 28 °C, debug log on; **bench only**). At room temperature the lamp lights; warm U2 with a fingertip or a hair dryer until `T=` passes 30 °C	ALERT pulls EN low within ~0.5 s: all LEDs off, `lp=--`. Below 28 °C the firmware re-initialises the LP5012 and the light returns. **Reflash the release build afterwards**	Check R2 (0402 on the TOP face, NE gap) and U2 pin 3
E18	LP5012 VCAP and IREF ground (round3 R3-E2: their ground returns to the LP5012's exposed pad are ~22–26 mm of pour)	4000 K at full, steady. AC-coupled scope probe on C4 (VCAP, bottom face) with a short ground spring to C4's GND pad; then E8's 1 Ω current reading over 1 min	VCAP ripple < 50 mV p-p with no bursts; channel current steady within ±1 %	Next board revision: EP vias + a short GND jumper for C4/R1 (layout note in round3.md)

4. Thermal (thermal review §7; thermocouples TC1 board top, TC2 LP5012, TC3 PLA skirt, TC4 PLA dome, TC7 cup)

#	Test	Duration	Pass
T1	Each mode at max level, 4.20 and 4.15 V: I_VLED cold and after 30 min	30 min	Within ±8 % of prediction (4000 K full ≈ 178 mA)
T2	Characterisation: derating disabled **on the bench only**, whites at 100 / 150 mA to steady state	~2 h	Gives the real board → PLA offset. Re-fit `T_DERATE_*` if it differs from ~14 K

#	Test	Duration	Pass
T3	Release firmware, 4000 K max, on USB, then on battery. TC2 directly on U1 (centre of the bottom face); log the debug `T=` alongside for 15 min at steady state (frozen-code check, R3-F6)	2 h each	TC3 ≤ 47 °C, TC4 ≤ 45 °C, TC1 ≤ 70 °C, TC2 ≤ 80 °C, TC7 ≤ 60 °C. Record TC2 − `T=` (the LP5012 runs hotter than the sensor: R3-E4) for T2's re-fit; the light must not "breathe" between 178 and 110 mA
T4	T3 in a 30–32 °C box	2 h	TC3 ≤ 50 °C
T5a	`pio run -e bench_t_nack` (sensor NACK) at full output	30 min	Output to 30 % within 3 s; TC3 ≤ 42 °C
T5b	`pio run -e bench_t_stuck` (frozen 25 °C reading)	2 h	Untrusted within 5 min (`T_FROZEN_MS`), current ≤ 110 mA, TC3 ≤ 47 °C
T6	Hardware thermostat: LEDs at 5 %, hot-air on the board underside to the MCP9800 ≥ 60 °C	10 min	LEDs off within 1 s of ALERT (LP5012 EN low); resume below 52 °C after the firmware re-init
T7	Creep: skirt flatness, lug height, skirt OD before and after T3+T4 and 10 × (2 h on / 1 h off)	1 day	Change ≤ 0.05 mm; bayonet still locks firmly

5. Optical and colour

#	Item	How	Pass	If not
O1	PLA transmission (T8a)	Print `coupon_litho_wedge` in the production PLA, backlight it per channel in a dark box	Note τ(t). Contrast visible from 0.9 to 2.6 mm	Change the lithophane range in `moonlib.py` (and re-run the optics model)
O2	Total flux (T8b)	12-direction illuminance at 1 m, 4000 K full, cold	≈ 37 lm ± 20 % (61 lm of LED flux × 0.61 escape, v0.2 head, head_v02.txt)	If < 30 lm, PLA absorption is higher than assumed: thin the base wall
O3	Page illuminance (T8c, F6)	Lux meter at 25 / 30 cm, 4000 K full, t = 0 and 30 min	20–30 lux expected	Feeds the next revision's reading-LED decision
O4	Uniformity (T8d)	RAW photos at fixed exposure: WW only, CW only, red only, all white	No visible 4-fold pattern (model max/min 1.028 one white / 1.021 all white / 1.037 RGB, docs/design/optics/head_v02.txt). The 45° collar band near the base (θ 30–45° from the bottom pole) reads darker and plain by design (B8: smooth and 2.6–3.5 mm thick)	Check the LED depth: emitters 1.2 mm below the opening (moon_shell B13). Deeper shades the low wall
O5	Flicker (T9; LP5012)	Photodiode + scope at levels 0, 3, 8, 15 for each mode; slow fades across brightness-register steps; phone slow-motion video	Content at the 29 kHz PWM and its ~3.6 kHz dither pattern only; no visible step when `lp_brightness()` changes	Turn PWM dithering off (`LP_CONFIG1` b2 = 0: 9-bit) or widen the hysteresis in `lp_brightness()`
O6	Colour per mode (T10, LED bins)	Spectrometer (or a phone colorimeter app as a rough check)	Whites within 3 SDCM / Duv ±0.003; candle Duv ±0.005	Put the measured xy and lumens into `LED` in `gen_modes.py` and regenerate
O7	Green chromaticity	Same as O6, green only	xy near (0.17, 0.70)	Update `LED["G"]` (520 nm usually sits greener)

6. Mechanical (from the parts' `VERIFY` dicts)

#	Item	Part	Check
M1	Gooseneck end thread M10×1, nut 14 AF × 3.0 mm; moon-end thread length ≥ 9 mm (jam nut + washer + floor + socket nut); any shoulder ≥ 12 mm on the end	`moon_base`, `pack_front`	Calipers on the delivered gooseneck. Change `NUT_AF`, `NUT_T`, `NECK_THREAD_D` in `moonlib.py` / `packlib.py`. A shoulder ≥ 12 mm can replace the jam nut (B16)
M12	Thread protrusion in the moon base (round2 R2-D3)	`moon_base`	With both nuts tight, the thread must not stand more than 0.5 mm above the socket nut's top; the board's underside is 4.5 mm above. Cut/file it to length **before** the cable goes in (handoff 05 §3)
M13	Battery lead path (round2 R2-D18)	`pack_back`, main board slot	Lead exit end and length on the real cell. The plug passes the 6.4 × 5.0 slot; the slack must lie flat in the bay's end gap without crossing the screw heads. If it doesn't, Claude adds a lead channel to `pack_back`
M14	LED-board jig (round2 R2-D12)	`jig_led_board`	The board drops in flush, and the bottom parts don't touch the relief floor
M2	Neck socket 9.3 mm for self-tapping into PETG	`pack_front`	Print `coupon_fit` first
M3	Printed slip fit 0.15 mm radial	all	`coupon_fit`
M4	Bayonet (v0.2, inside out; round 3): the skirt slides over the cup, the lugs enter its slots, a **clockwise** 30° turn (seen from the top) pulls the moon down onto the PETG ring, preloading it by `LOCK_OFFSET` 0.15 mm + (disc − 1.0); the ring's roll over the cup's land is the spring	`moon_shell`, `moon_base`	`coupon_bayonet_cup` (PETG) + `coupon_bayonet_skirt` (the moon's PLA), with 18 mm discs 0.9, 1.0 and 1.1 mm thick. Pass: locks by hand without forcing, no whitening at the groove ends, no rattle. Tune `LOCK_OFFSET` in `moonlib.py` (lower = looser) in 0.05 mm steps and reprint both halves
M5	Ring preload: board must not rattle with the moon locked (ring 0.8 mm, ID 16.4, at 0.12 mm layers; clamp cone on its 45° chamfer)	`moon_ring`	Tune `LOCK_OFFSET` (M4). No shims: they add rigid interference on top of the spring's travel
M6	LED board thickness 1.0 mm as ordered	`moon_base`	Calipers on the delivered board
M7	Battery 64.0 × 35.5 × 5.3 mm (listing)	`pack_back`	Calipers on the delivered cell; bay is 36.0 × 64.5 × 5.6 + 0.6 swell
M8	USB-C receptacle centre 1.63 mm above the PCB; PCM12 knob 0.9 mm above, 1.0 mm past the edge; J2 header 5.5 mm tall (JST), under a 0.8 mm face pocket; your USB-C cables seat fully in the 13.4 × 7.6 window	`pack_front`	Calipers on the assembled main board before printing the shell; try your cables in the printed shell
M9	Clip force ~5 N at 3 mm	`pack_clip`	Holds on a hardback and a paperback without marking them
M10	Button caps 7.0 mm in 7.4 mm holes	`button_caps`	Slide freely; sand if tight
M11	Lithophane 0.9–2.6 mm reads well at 0.12 mm layers	`moon_shell`	First print, lit and unlit